

DBMS PRACTICE 1

SECTION 1 : DDL (DATA DEFINITION LANGUAGE)

DDL COMMANDS:

1. CREATE
2. ALTER
3. DROP
4. TRUNCATE
5. RENAME

QUESTION 1 - CREATE TABLE

```
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| my_database        |
| mysql              |
| performance_schema |
| student_marks      |
| sys                |
+-----+
6 rows in set (0.00 sec)

mysql> create database bank_accounts;
Query OK, 1 row affected (0.01 sec)

mysql> use bank_accounts;
Database changed
mysql> create table bank_transactions (
-> txn_id int primary key,
-> customer_name varchar(50),
-> branch_name varchar(50),
-> transactions_type varchar(50),
-> amount decimal(10,2),
-> transaction_date date
-> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> alter table bank_transactions
-> modify customer_name varchar(100);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> |
```

QUESTION 2 - ALTER TABLE ADD COLUMN

```
mysql> alter table bank_transactions
    -> add account_no varchar(20);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> |
```

QUESTION 3 - ALTER TABLE MODIFY COLUMN

QUESTION 4 - RENAME TABLE

```
mysql> rename table bank_transactions
    -> to customer_transactions;
Query OK, 0 rows affected (0.02 sec)

mysql> |
```

QUESTION 5 - TRUNCATE TABLE

```
mysql> rename table customer_transactions
    -> to bank_transactions;
Query OK, 0 rows affected (0.04 sec)

mysql> truncate table bank_transactions;
Query OK, 0 rows affected (0.07 sec)

mysql> |
```

QUESTION 6 - DROP TABLE

```
mysql> create table bank_backup (
    -> id int
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> drop table bank_backup;
Query OK, 0 rows affected (0.04 sec)

mysql> |
```

SECTION 2 : DML (DATA MANIPULATION LANGUAGE)

DML COMMANDS:

1. INSERT
2. UPDATE
3. DELETE
4. SELECT

STEP 1 : INSERT SAMPLE DATA

```
mysql> insert into bank_transactions values
-> (101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05', NULL),
-> (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06', NULL),
-> (103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08', NULL),
-> (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10', NULL),
-> (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11', NULL),
-> (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12', NULL),
-> (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13', NULL),
-> (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14', NULL),
-> (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15', NULL),
-> (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16', NULL);
Query OK, 10 rows affected (0.03 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> |
```

QUESTION 7 – INSERT

```
mysql> insert into bank_transactions values
-> (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18', NULL);
Query OK, 1 row affected (0.02 sec)
```

```
mysql> select*from bank_transactions;
```

txn_id	customer_name	branch_name	transactions_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
111	Venu	Vizag	Deposit	7000.00	2024-01-18	NULL

```
11 rows in set (0.00 sec)
```

```
mysql> |
```

QUESTION 8 – UPDATE

```
mysql> update bank_transactions
-> set amount = 5000
-> where txn_id = 105;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select*from bank_transactions
-> where txn_id = 105;
```

txn_id	customer_name	branch_name	transactions_type	amount	transaction_date	account_no
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL

1 row in set (0.00 sec)

```
mysql> |
```

QUESTION 9 – DELETE

```
mysql> delete from bank_transactions
-> where txn_id = 111;
Query OK, 1 row affected (0.03 sec)
```

```
mysql> select*from bank_transactions;
```

txn_id	customer_name	branch_name	transactions_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

10 rows in set (0.00 sec)

```
mysql> |
```

QUESTION 10 - SELECT

```
mysql> select*from bank_transactions;
```

txn_id	customer_name	branch_name	transactions_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

```
10 rows in set (0.00 sec)
```

```
mysql> |
```

QUESTION 11 SELECT WITH WHERE

```
mysql> select*from bank_transactions
-> where transactions_type = 'Deposit';
```

txn_id	customer_name	branch_name	transactions_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

6 rows in set (0.00 sec)

```
mysql> |
```

QUESTION 12 - SELECT WITH ORDER BY

```
mysql> select*from bank_transactions
-> order by amount desc;
```

txn_id	customer_name	branch_name	transactions_type	amount	transaction_date	account_no
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL

10 rows in set (0.00 sec)

```
mysql> |
```

SECTION 3 : DCL (DATA CONTROL LANGUAGE)

DCL COMMANDS:

1. GRANT
2. REVOKE

QUESTION 13 - GRANT SELECT

```
mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
```

Grants for Auditor1@localhost
GRANT USAGE ON *.* TO 'Auditor1'@'localhost'
GRANT SELECT ON 'bank_accounts'.'bank_transactions' TO 'Auditor1'@'localhost'

2 rows in set (0.00 sec)

```
mysql> |
```

QUESTION 14 - GRANT ALL PRIVILEGES

```
mysql> create user 'branchmanager'@'localhost'
-> identified by 'manager123';
Query OK, 0 rows affected (0.03 sec)

mysql> grant all privileges on bank_transactions
-> to 'branchmanager'@'localhost';
Query OK, 0 rows affected (0.03 sec)

mysql> show grants for 'branchmanager'@'localhost';
+-----+
| Grants for branchmanager@localhost |
+-----+
| GRANT USAGE ON *.* TO `branchmanager`@`localhost` |
| GRANT ALL PRIVILEGES ON `bank_accounts`.`bank_transactions` TO `branchmanager`@`localhost` |
+-----+
2 rows in set (0.00 sec)

mysql> |
```

QUESTION 15 - REVOKE SELECT

```
mysql> revoke select on bank_transactions
-> from 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.03 sec)

mysql> show grants for 'Auditor1'@'localhost';
+-----+
| Grants for Auditor1@localhost |
+-----+
| GRANT USAGE ON *.* TO `Auditor1`@`localhost` |
+-----+
1 row in set (0.00 sec)

mysql> |
```

QUESTION 16 REVOKE ALL

```
mysql> revoke all privileges on bank_transactions
-> from 'branchmanager'@'localhost';
Query OK, 0 rows affected (0.03 sec)

mysql> show grants for 'branchmanager'@'localhost';
+-----+
| Grants for branchmanager@localhost |
+-----+
| GRANT USAGE ON *.* TO `branchmanager`@`localhost` |
+-----+
1 row in set (0.00 sec)

mysql> |
```


SECTION 4 : TCL (TRANSACTION CONTROL LANGUAGE)

TCL COMMANDS:

1. COMMIT
2. SAVEPOINT
3. ROLLBACK

QUESTION 17 – COMMIT

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> update bank_transactions
  -> set amount = 6000
  -> where txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> commit;
Query OK, 0 rows affected (0.02 sec)

mysql> select*from bank_transactions
  -> where txn_id = 101;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transactions_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
|    101 | Ravi          | Hyderabad  | Deposit          | 6000.00 | 2024-01-05      | NULL       |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

QUESTION 18 – SAVEPOINT

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> savepoint before_update;
Query OK, 0 rows affected (0.00 sec)

mysql> |
```


QUESTION 19 - ROLLBACK TO SAVEPOINT

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> savepoint before_update;
Query OK, 0 rows affected (0.00 sec)

mysql> update bank_transactions
    -> set amount = 99999
    -> where txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> rollback to before_update;
Query OK, 0 rows affected (0.02 sec)

mysql> commit;
Query OK, 0 rows affected (0.00 sec)

mysql> select*from bank_transactions
    -> where txn_id = 102;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transactions_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

QUESTION 20 FULL TCL EXAMPLE

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> update bank_transactions
-> set amount = 7000
-> where txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> savepoint sp1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> update bank_transactions
-> set amount = 9000
-> where txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> rollback to sp1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> commit;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select*from bank_transactions
-> where txn_id in (101, 102);
```

txn_id	customer_name	branch_name	transactions_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	7000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL

```
2 rows in set (0.00 sec)
```

```
mysql> |
```